INCH-POUND

MIL-DTL-24640/15D
22 August 2002

SUPERSEDING
MIL-C-24640/15C
26 June 1995

DETAIL SPECIFICATION SHEET

CABLE, ELECTRIC, 600 VOLTS, LIGHT-WEIGHT, MULTI-PAIR, AWG 22, TYPE 2XSAW (INCLUDING VARIATION TYPE 2XSAOW)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of MIL-DTL-24640 listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation.

Construction (Watertight)

First - 7 strands, tin-coated copper conductor, AWG size and dimensions in accordance with Table I.

Second - Insulation material and basic construction requirements shall be in accordance with MIL-W-81044/12 and tested in accordance with the component wire ratings and requirements listed on this slash sheet. Quality conformance testing shall be in accordance MIL-W-81044, prior to cabling.

Third - Two conductors, one black and one white, twisted together. The pitch shall be 1¼ ± ¼ inch.

Fourth - Binder tape applied helically with overlap, at manufacturer's option.

Fifth - Braided shield of # 38 AWG tin coated copper strands. Braid coverage shall be no less than 90 percent determined in accordance with WC-27500.

Sixth - Shield insulation of two polyester tapes or a polyester tape and extruded jacket. Identification codes applied using Method 2.

Seventh - The required number of shielded, twisted pairs shall be cabled together. Cabling sequence shall be consecutive, starting with number 1 from center outward. Fillers may be used as necessary, to form a round cable.

Eighth - Binder tape applied helically with overlap.

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T23:44Z
Check the source to verify that this is the current version before use.

CONSTRUCTION VARIANT FOR TYPE 2XSAW

Ninth - Cross-linked polyolefin jacket. (See Table II for thickness).

CONSTRUCTION VARIANT FOR TYPE 2XSAOW

Ninth - Braided shield, tin coated copper strands. Braid must be constructed to meet Surface Transfer Impedance requirements as listed herein.
Tenth - Binder tape applied helically with overlap.
Eleventh -Cross-linked polyolefin jacket. (See Table III for thickness).

TABLE I. Component wire details.

Watertight conductor			Finished wire			
Wire size (AWG)	Stranding (no. x AWG)	Diameter nominal (inches)	Diameter (inches)	Test load life cycle accelerated aging tests (lbs. ± 3%)	Low temp. at -40° C	
					Mandrel diameter (inches)	Test load (lbs)
22	7 x 30	0.030	0.047 ± 0.003	0.190	0.75	1.0

TABLE II. Construction details.

PIN M24640/15	Type number	Number of pairs	Jacket thickness (inch) (min)	Cable diameter (inch)		Cable weight (pounds per 1000 feet) (max)
				(min)	(max)	
-01UN	2XSAW-3	3	0.041	0.368	0.396	115.0
-02UN	2XSAW-7	7	.041	.461	.497	195.0
-03UN	2XSAW-14	14	.041	.641	.691	390.0

TABLE III. Construction details - overall shield.

PIN M24640/15	Type number	Number of pairs	Diameter over shield (inch) (nom)	Jacket thickness (inch) (min)	Cable diameter (inch)		Cable weight (pounds per 1000 feet) (max)
					(min)	(max)	
-01UO	2XSAOW-3	3	0.291	0.041	0.405	0.437	160.0
-02UO	2XSAOW-7	7	.400	.044	.510	.550	260.0
-03UO	2XSAOW-10	10	.533	.044	.631	.681	385.0
-04UO	2XSAOW-14	14	.585	.047	.689	.743	465.0
-05UO	2XSAOW-19	19	.656	.047	.757	.817	627.0
-06UO	2XSAOW-24	24	.791	.047	.884	.952	800.0
-07UO	2XSAOW-30	30	.843	.050	.941	1.02	910.0
-08UO	2XSAOW-37	37	.924	.050	1.01	1.09	1050.0

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T23:44Z
Check the source to verify that this is the current version before use.

REQUIREMENTS:

Qualification required.
Construction details: See Tables I, II, and III.

Component wire: The component wire shall be per MIL-W-81044 and shall be tested in accordance with the following wire ratings and additional requirements.

Temperature rating: 135°C (275°F) maximum conductor temperature
Voltage rating: 600 volts (rms) at sea level.
Accelerated aging: Oven temperature, 250 ± 2°C (482 ± 3.6°F) for 6 hours.
Low temp.-cold bend: -40°C (-40°F) for 4 hours
Blocking: 150 ± 2°C (302 ± 3.6°F)
Color: In accordance with MIL-STD-104, class 1.
Flammability: 30 seconds (max); 3.0 inches (max); no flaming of tissue paper.
Humidity resistance: 5000 megohms for 1000 feet, minimum insulation resistance after humidity exposure.
Identification, striping or banding durability: 125 cycles (250 strokes) (min), 500 gram weight.
Impulse dielectric test:
    Primary insulation (when tested in lieu of spark test):
        6.0 kilovolts (peak) 100 percent test.
    Finished wire: 8.0 kilovolts (peak), or 3 kHz spark test
    5.5 kV, 100 percent test.
Insulation resistance: 5000 megohms for 1000 feet (min).
Life cycle: Oven temperature, 200 ± 2°C (392 ± 3.6°F) for 120 hours.
Physical properties of primary insulation:
    Tensile strength, 2500 lb/in^2 (min).
    Elongation, 150 percent (min).
Shrinkage: 0.125 inch maximum at 250 ± 2°C (482 ± 2°F).
Spark test primary insulation: 1500 volts (rms), 60 Hz or 3 KHz, 100 percent test.
Surface resistance: 500 megohms inches (min), initial and final readings.
Thermal shock: Oven temperature, 150 ± 2°C (302 ± 3.6°F).
    Maximum change in measurement: 0.060 inch.
Wet dielectric test: 2500 volts (rms).
Wrap test: Mandrel wrap test required.

EXAMINATION AND TESTS:

Basic electrical:	Requirements:
Conductor Resistance - ohms/1000 feet at 25°C, maximum	17.89
Voltage withstand - volts, root mean square, minimum	
Conductor to conductor and shield	2000

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T23:44Z
Check the source to verify that this is the current version before use.

Conductor to shield 1000
Component shield to component shield 200
Component shield to overall shield (2XSAOW only) 500
Insulation resistance – megohms-1000 feet, minimum
Conductor to conductor and shield 2500
Shield to shield (component shield only) 100
Shield to shield (component shield to overall shield (2XSAOW only) 100
Conductor and shield continuity ..... No failure
Jacket flaws (type 2XSAOW only) ..... No failure

Group A:
Visual and dimensional ..... No failure
Capacitance:
Mutual capacitance - at 1 MHz, picofarads/foot (pF/ft), maximum 55
Capacitance unbalance - percent, maximum 15
Characteristic impedance - at 1 MHz, ohms ..... 33 min, 72 max
Watertightness (see Table IX of MIL-DTL-24640 for
limits of water leakage) ..... No failure
Weight ..... No failure

Group B:
Cold bending ..... No failure
Drip ..... No failure
Cross link proof test (percent maximum)
Jacket 50
Tear - pounds per inch thickness, minimum 35

Physicals (unaged)
Jacket
Tensile strength - lb/in^2, minimum 1300
Elongation - percent, minimum 160
Attenuation - at 3 MHz, dB/100 feet, maximum ..... Not applicable
Gas flame (1 hour) ..... Not applicable

Group C:
Physicals (aged)
Jacket
Air oven
Tensile strength - percent of unaged, minimum 60
Elongation - percent of unaged, minimum 60
Permanence of printing (conductor) - cycles, minimum ..... Not applicable
Permanence of printing (jacket) - cycles, minimum 125
Shield - conformance to material, construction and coverage ..... No failure
Heat distortion - percent of unaged, maximum 30
Cable sealant removability ..... No failure

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T23:44Z
Check the source to verify that this is the current version before use.

Shrinkage ..... No failure
Mutual inductance - dB/1000 feet, maximum ..... Not applicable
Surface transfer impedance (type 2XSAOW only)
    Milliohms per meter, maximum 700
    EMP response, dB, minimum 60
Pulse response time - ms/500 feet
    Rise time, maximum ..... Not applicable
    Decay time, maximum ..... Not applicable

QUALIFICATION INSPECTION:

Qualification inspection shall include basic electrical, all of groups A, B, C and D, plus the following:

Flame propagation ..... No failure
Aging and compatibility (cable) ..... No failure
Acid gas equivalent - percent, maximum
    Jacket 2
    Fillers 2
    Insulation 18
Halogen content - percent, maximum
    Jacket 0.2
    Fillers 0.2
Hot oil immersion
    Tensile strength - percent of unaged, minimum 50
    Elongation - percent of unaged minimum 50
Smoke index, maximum
    Jacket 25
    Fillers 45
    Insulation 10
Toxicity index, maximum
    Jacket 5
    Fillers 5
    Insulation 1.5
Durometer (jacket) - (type A) hardness, minimum 80
Weathering (jacket) ..... No failure
Fungus Resistance ..... No failure
Abrasion resistance ..... No failure
Gas Flame (3 hour) ..... Not applicable
Component wire ..... No failure

Unit ordering length: 3 & 14 500 feet nominal.
19 - 37 250 feet nominal

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T23:44Z
Check the source to verify that this is the current version before use.

Custodians:
Army - MI
Navy - SH

Preparing Activity:
Navy - SH
(Project 6145-2307-015)

Review Activities:
Army - AR, AV, CR, CR4
Navy - CG, EC
DLA - IS

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T23:44Z
Check the source to verify that this is the current version before use.